

Ford GoBike System 2018

Exploratory Data Analysis Report

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Objective: Understand user behavior, temporal patterns, station utilization, and derive actionable business recommendations.

1. Executive Summary

The 2018 Ford GoBike dataset contains **1,863,721** cleaned trip records. Analysis reveals a strongly commuter-driven service dominated by subscribers, with clear opportunities to convert casual customers and optimize operations.

Key Highlights:

- Subscribers represent most rides with predictable commute patterns.
 - Customers take longer leisure trips, especially on weekends.
 - Significant seasonal variation and station concentration observed.
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2. Dataset Overview

- **Total Records:** 1,863,721
- **Time Period:** January – December 2018
- **Key Features:** Trip duration, start/end time, station info, user type, member demographics.

[INSERT DATA SUMMARY TABLE HERE]

3. Data Cleaning & Feature Engineering

Actions Performed:

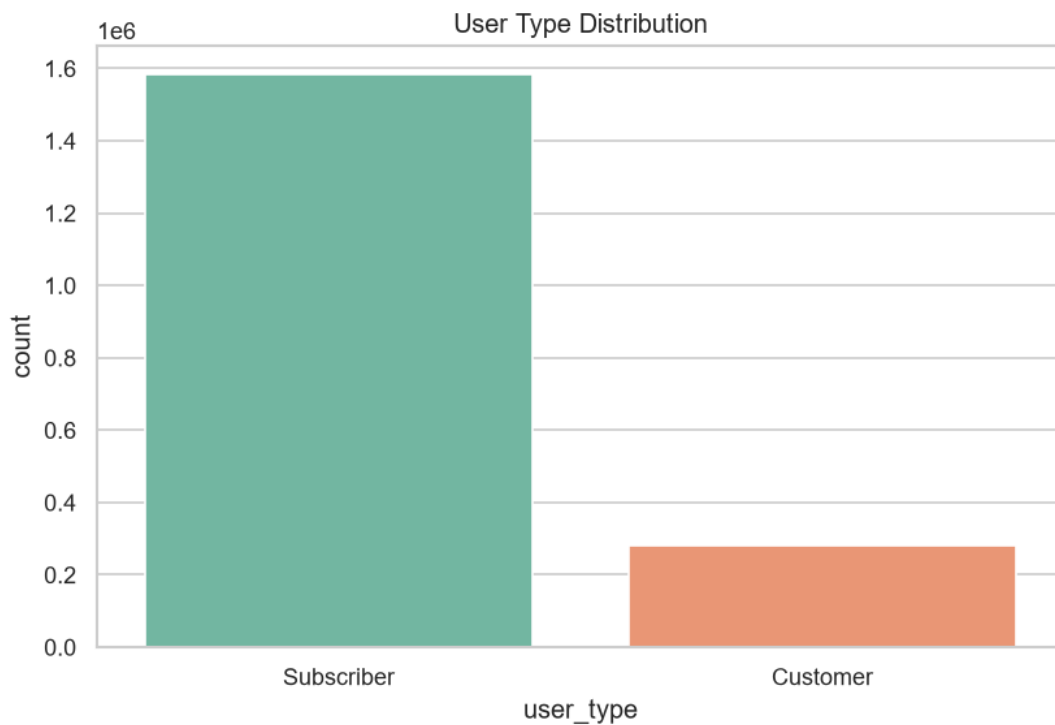
- Combined 12 monthly CSV files

- Parsed datetime columns
- Removed duplicates and implausible durations
- Created features: duration_min, start_hour, day_of_week, is_weekend, age
- Handled missing values in gender and birth year

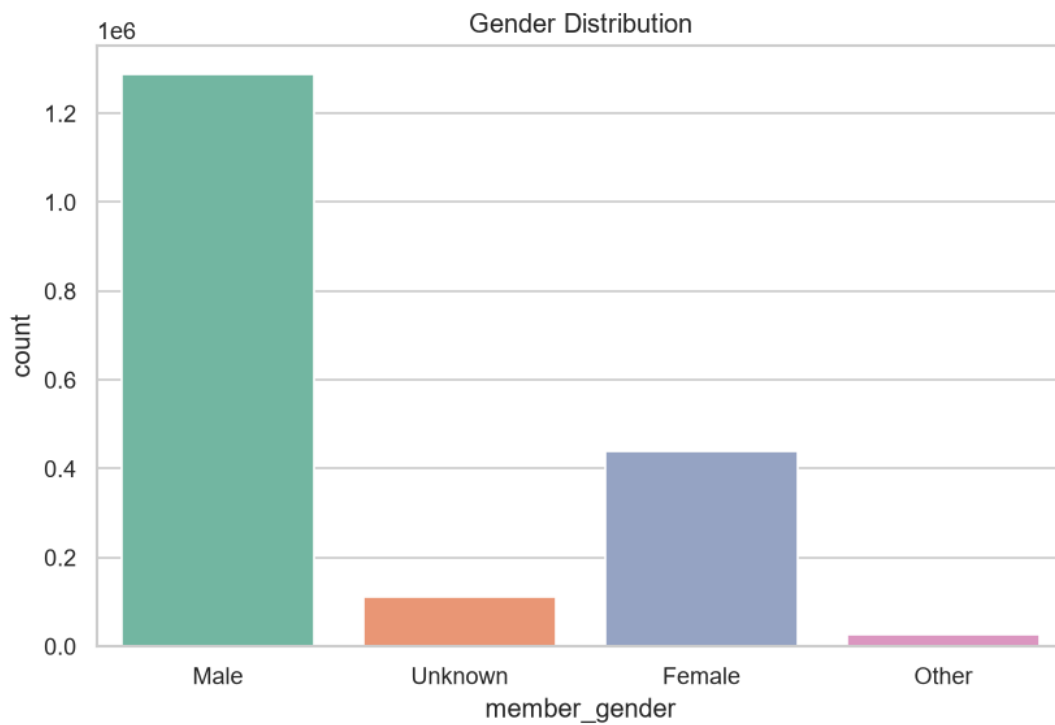
Final Clean Shape: 1,863,721 rows

4. Exploratory Data Analysis

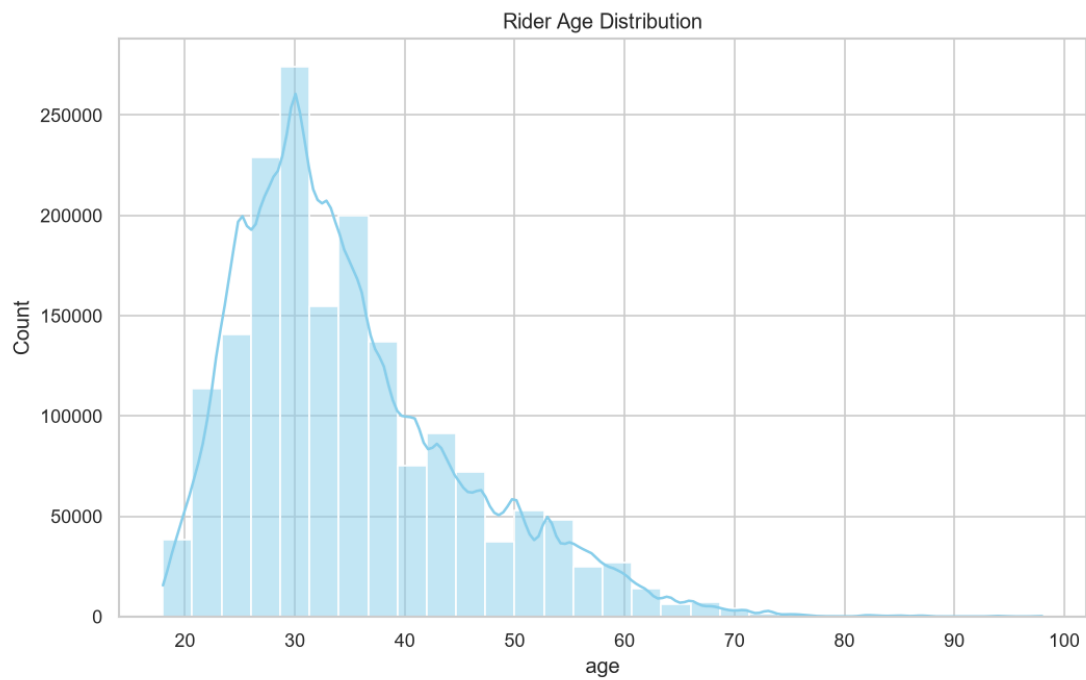
4.1 User Segmentation



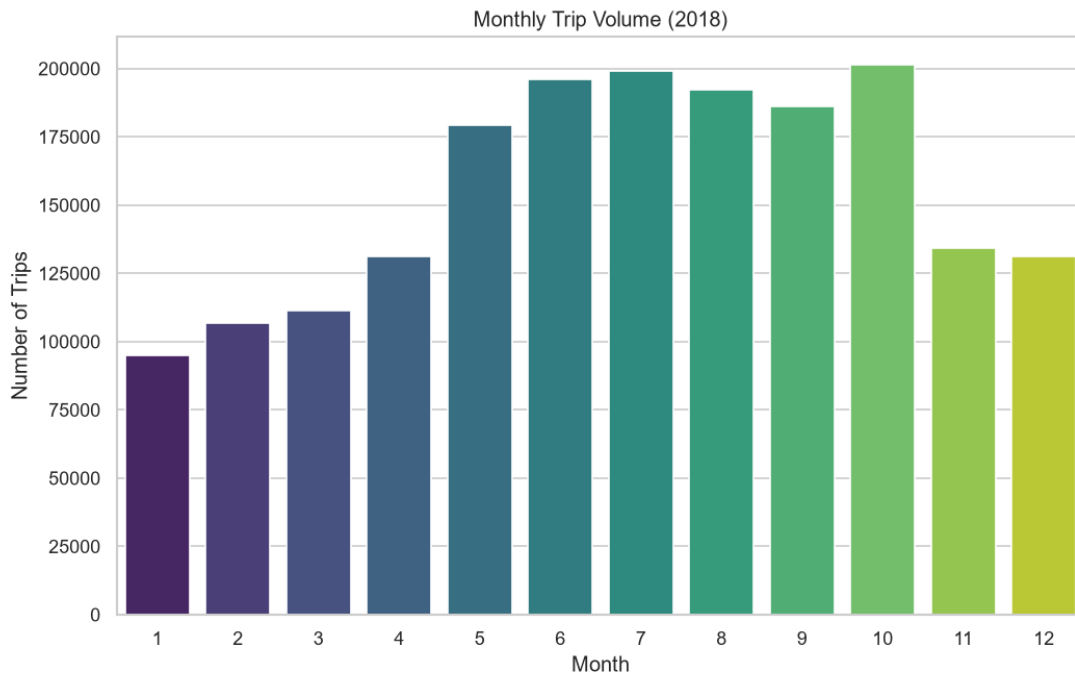
4.2 Gender Distribution



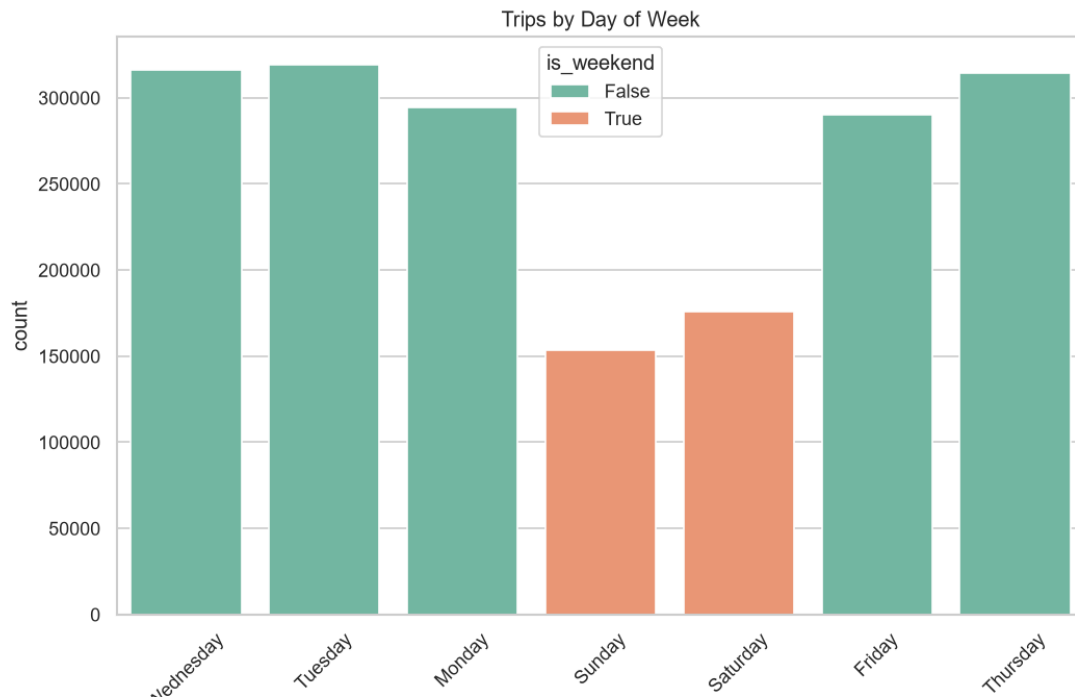
4.3 Age Distribution



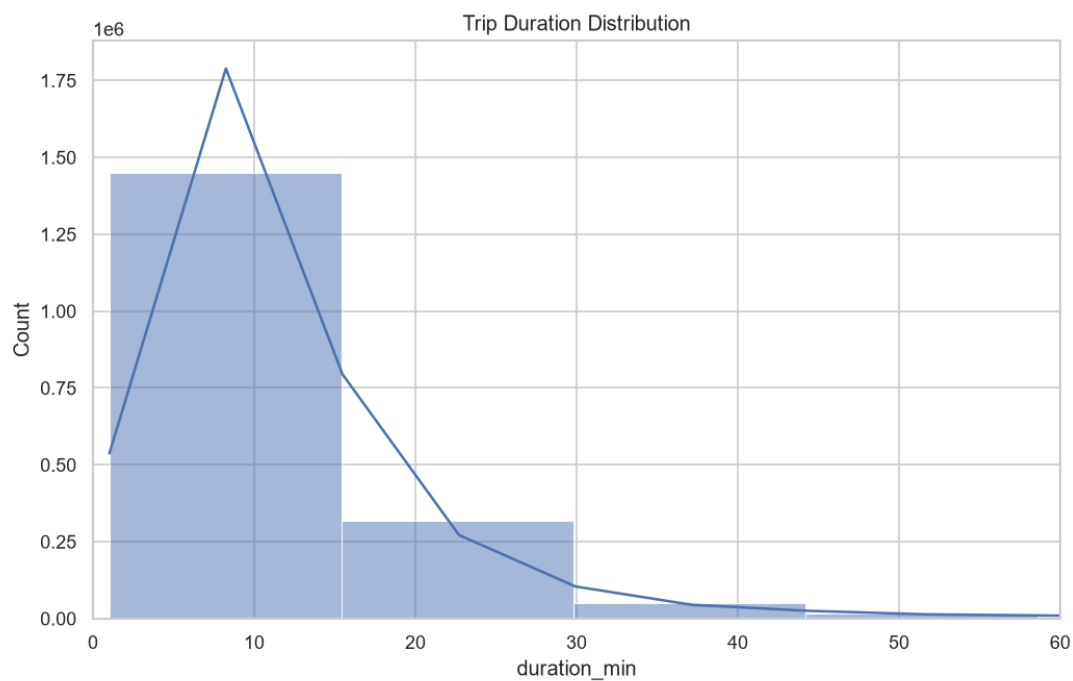
4.4 Monthly Trip Volume



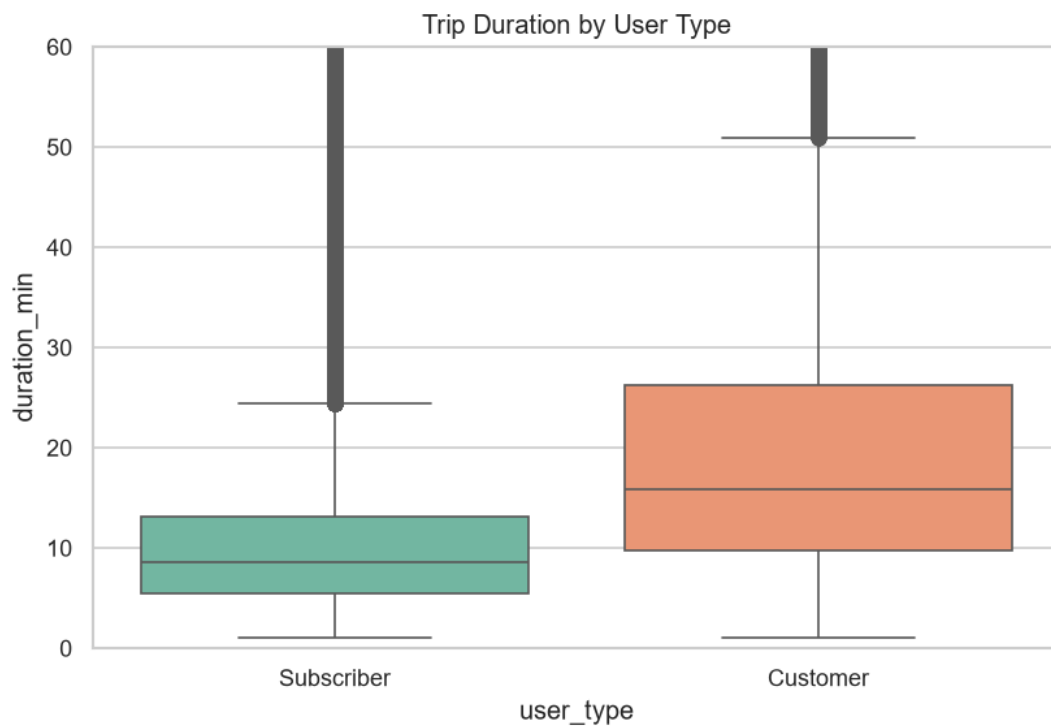
4.5 Hourly Usage Pattern by User Type



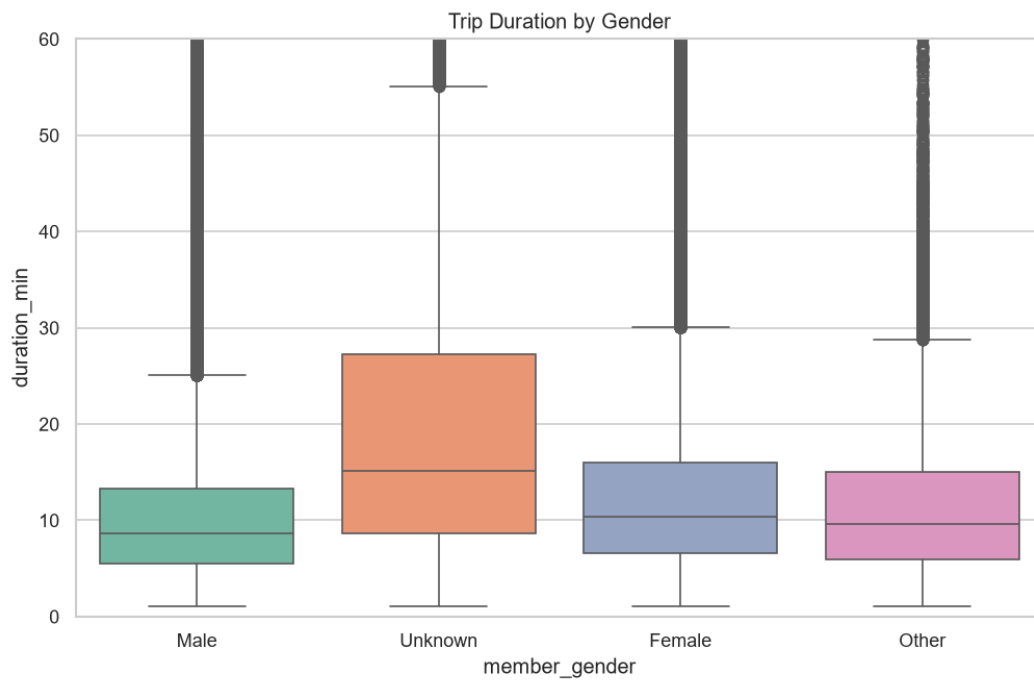
4.6 Trip Duration Distribution



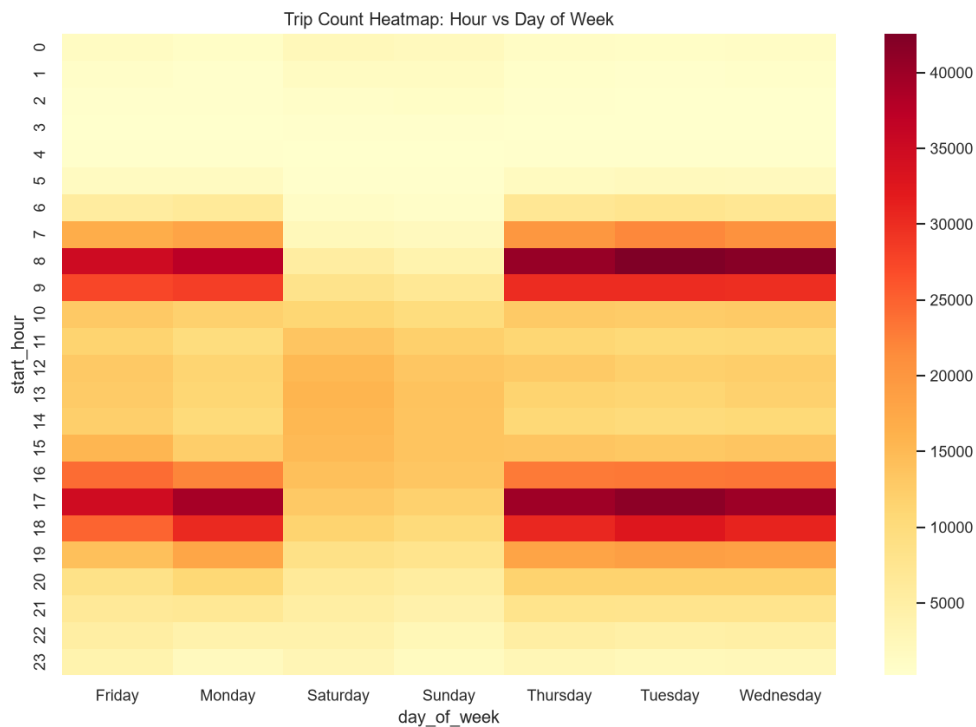
4.7 Trip Duration by User Type



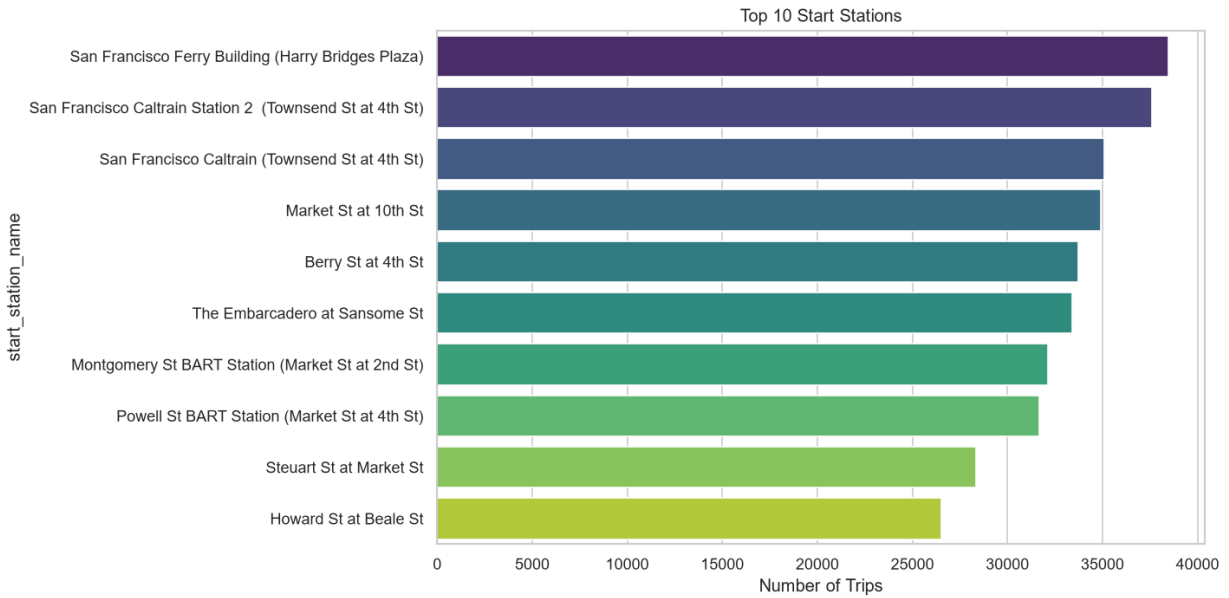
4.8 Trip Duration by Gender



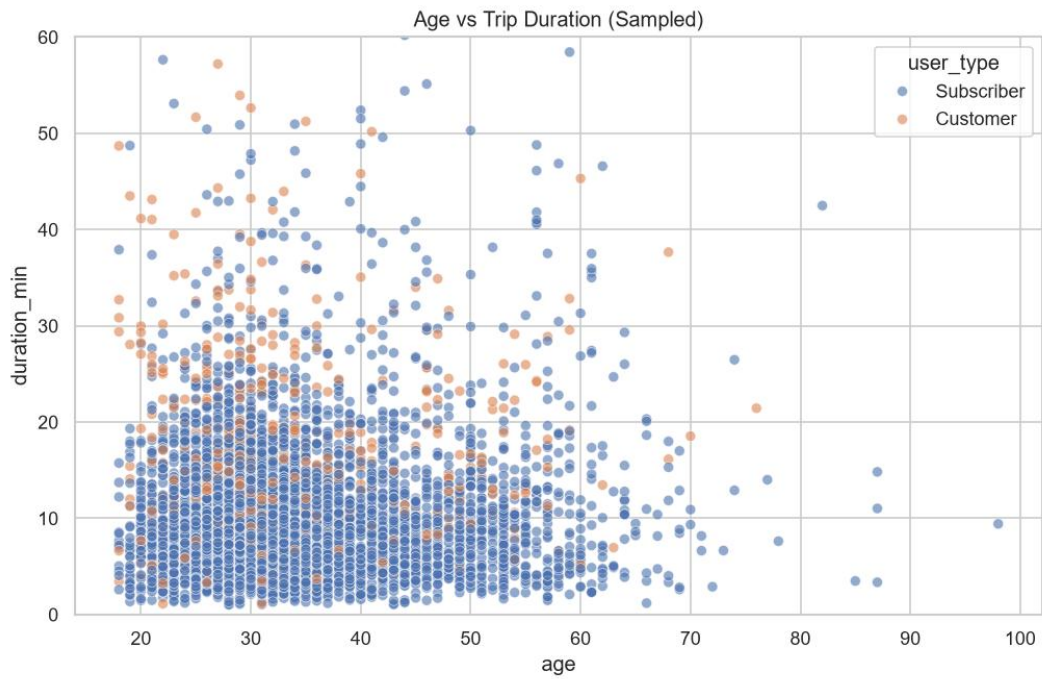
4.9 Top 10 Start Stations



4.10 Age vs Trip Duration



4.11 Hour vs Day of Week Heatmap



4.12 Weekend vs Weekday Duration



5. Key Business Insights

1. **Commuter Dominance:** Subscribers drive peak-hour usage.
2. **Casual User Opportunity:** Customers show longer trips and weekend preference.
3. **Demographic Skew:** Young males (25-34) are core users.
4. **Operational Bottlenecks:** Few stations handle majority of traffic.
5. **Seasonality:** Winter usage drops significantly.

6. Strategic Recommendations

Marketing:

- Implement post-ride subscription prompts for customers.
- Launch targeted campaigns for female riders focusing on safety.

Operations:

- Optimize rebalancing schedules by day type.
- Prioritize infrastructure upgrades at top stations.

- Scale fleet/staff seasonally.

Product Development:

- Introduce flexible weekend passes.
 - Explore corporate subscription packages.
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7. Conclusion

This analysis provides a solid foundation for data-driven decision making to increase ridership, improve user retention, and optimize operational efficiency.